V2 fiscal model — scenario tables

Policy Director · built 12 October 2026 · internal, pre-decisional · MOO-100

This document exists because the model must be managed as a document. Every number that the interactive model produces when you move a slider is printed here, across the full range of each slider. Nothing in the HTML model is absent from these tables. The constants, the formulas and the pool counts are identical — this is the same model, computed and printed instead of computed and drawn.

Governed by **V2-goal-set** and by the six board decisions of 12 October 2026 recorded in **V2-decisions-adopted**. All six are applied. Decisions 3 to 6 confirmed the reading these tables were already computed on — the purchasable Bridge screen, the non-hazardous State Employees scope, both Plan 2 findings, and no transfer mechanism — so **no figure in this document changed when they were taken.** Built to **G10** (ranges, never a single forecast), **G11** (three separate ledgers), **G12** (Bridge and Incentive never averaged together) and **G13** (no claim of minimal impact, actuarial neutrality or self-funding).

Nothing here is a VRS actuarial result. Ledger 1 stops at the member payment, which is verified, and expresses the trust-fund subsidy as a multiple **k** of that payment, because only VRS can supply the actuarial-equivalent cost.

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1. The inputs, and which of them are verified

Six quantities move in this model. Three are verified facts and never move; three are judgments and are printed across their whole range in every table below.

Table 1 — fixed inputs. These do not move.

Input	Value	Source
State Plan 1 / Plan 2 employees	28,994	VRS Annual Actuarial Valuation, 30 June 2025, p. 50 — age × total-service table for all 81,274 active State Employees, with statutory tier boundaries applied. Plan 1 20,872 · Plan 2 8,122.
Member price, Plan 1	12.50% of salary per year purchased	VRS approximate-normal-cost rate chart, effective 1 July 2025
Member price, Plan 2	10.74% of salary per year purchased	VRS approximate-normal-cost rate chart, effective 1 July 2025
Incentive pool	6,254 — Plan 1: 5,928 · Plan 2: 326	Derived from the same valuation table — members already past an unreduced-retirement threshold
Bridge pool, purchasable screen	2,396 — Plan 1 only	Members who can actually reach a threshold by buying up to 48 months. See section 7.
Bridge pool, G9 screen	5,177 — Plan 1: 4,635 · Plan 2: 542	G9 as written — within four years of a threshold, whether or not purchased service can get them there
Employer on-cost multiplier	1.1223	Group life 0.0434 + admin 0.0024 + FICA 0.0765. Excludes the 6.49-point amortization rate — see section 6.
Benefit multiplier	Plan 1: 1.70% · Plan 2: 1.65%	Virginia Code § 51.1-155

Table 2 — the three judgment dials. Every table below is printed across these.

Dial	Low	Mid	High	What would close it
Veteran share of the eligible pool	6%	8%	10%	G24 — DHRM veteran-status field. It currently reads "unknown" for most of the workforce; DHRM records 6%, labour-force parity is 8.1%.
Share with qualifying purchasable service	35%	50%	65%	G24 — VRS has no published figure. This is the weakest number in the model.
Take-up among those eligible	5%	12%	25%	Board decision of 12 October 2026. Only Virginia pilot experience closes it, which is G24's purpose.

A fourth dial, **k**, is not a judgment we are entitled to make — it is the ratio of actuarial-equivalent cost to approximate normal cost, and VRS owns it. Section 5 prints the subsidy at k = 1.5, 2.0, 2.5 and 3.0. Two further dials, replacement salary and positions not refilled, belong to Ledger 2 and are printed in full in section 6.

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2. The eligibility funnel

Five stages, top to bottom. Stage 1 is verified, stage 2 is derived from the published table, stages 3 to 5 are the three dials.

Table 3 — funnel, all three cases. "Retained" is the share of the stage above.

Case	Stage	Members	Retained
Low (6/35/5)	1. State Plan 1 / Plan 2 employees	28,994	—
	2. Pathway-eligible	8,650	29.8%
	3. Veterans among them	519	6.0%
	4. With qualifying purchasable service	182	35.0%
	5. Participants	9.1	5.0%
Mid (8/50/12)	1. State Plan 1 / Plan 2 employees	28,994	—
	2. Pathway-eligible	8,650	29.8%
	3. Veterans among them	692	8.0%
	4. With qualifying purchasable service	346	50.0%
	5. Participants	42	12.0%
High (10/65/25)	1. State Plan 1 / Plan 2 employees	28,994	—
	2. Pathway-eligible	8,650	29.8%
	3. Veterans among them	865	10.0%
	4. With qualifying purchasable service	562	65.0%
	5. Participants	141	25.0%

The pathway-eligible pool is 8,650 under the purchasable screen — 30% of all Plan 1/Plan 2 state employees. Everything after that stage is multiplicative, which is why the participant range is wide.

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3. Reach — the full 27-cell participant grid

This is the single most asked-for number and it has no point estimate. Below is every combination of the three dials. Participants are for the whole four-year pilot, not per year.

Table 4 — participants over the four-year pilot, every dial combination. Purchasable Bridge screen.

Veteran share	Qualifying share	Take-up	Participants	— Incentive	— Bridge	Member payments
6%	35%	5%	9.1	6.6	2.5	$121k
6%	35%	12%	22	15.8	6.0	$291k
6%	35%	25%	45	33	12.6	$607k
6%	50%	5%	13.0	9.4	3.6	$173k
6%	50%	12%	31	23	8.6	$416k
6%	50%	25%	65	47	18.0	$867k
6%	65%	5%	16.9	12.2	4.7	$225k
6%	65%	12%	40	29	11.2	$541k
6%	65%	25%	84	61	23	$1.13M
8%	35%	5%	12.1	8.8	3.4	$162k
8%	35%	12%	29	21	8.1	$388k
8%	35%	25%	61	44	16.8	$809k
8%	50%	5%	17.3	12.5	4.8	$231k
8%	50%	12%	42	30	11.5	$555k
8%	50%	25%	86	63	24	$1.16M
8%	65%	5%	22	16.3	6.2	$301k
8%	65%	12%	54	39	15.0	$721k
8%	65%	25%	112	81	31	$1.50M
10%	35%	5%	15.1	10.9	4.2	$202k
10%	35%	12%	36	26	10.1	$486k
10%	35%	25%	76	55	21	$1.01M
10%	50%	5%	22	15.6	6.0	$289k
10%	50%	12%	52	38	14.4	$694k
10%	50%	25%	108	78	30	$1.45M
10%	65%	5%	28	20	7.8	$376k
10%	65%	12%	67	49	18.7	$902k
10%	65%	25%	141	102	39	$1.88M

Range 9 to 141. The board's reach decision fixes take-up at 5 / 12 / 25% and holds the other two dials at mid, which gives **17 / 42 / 86**. Those three are the figures to quote; the grid shows what happens if the other two dials are wrong, and they can each move the answer by more than take-up does.

Table 5 — the board case. Veteran share 8%, qualifying share 50%, take-up varied.

Take-up	Participants	— Incentive	— Bridge	Member payments	Average salary
5%	17.3	12.5	4.8	$231k	$84,329
12%	42	30	11.5	$555k	$84,329
25%	86	63	24	$1.16M	$84,329

4. Pathway split — G12 requires these never be averaged

The two pathways have different economics, different sizes and, critically, different plan composition. The Incentive is roughly 72% of participants in every case.

Table 6 — Incentive against Bridge, by case.

Case	Pathway	Pool	Participants	Share	Member payments	Months at normal cost
Low	Incentive	6,254	6.6	72%	$69k	up to 12 months
Low	Bridge	2,396	2.5	28%	$52k	up to 48 months, only what is needed
Mid	Incentive	6,254	30	72%	$316k	up to 12 months
Mid	Bridge	2,396	11.5	28%	$239k	up to 48 months, only what is needed
High	Incentive	6,254	102	72%	$1.07M	up to 12 months
High	Bridge	2,396	39	28%	$810k	up to 48 months, only what is needed

Expected Plan 2 Bridge participation is zero for the whole pilot, and that is structural, not small. A Plan 2 member holds at most 15.5 years of service, so the Rule of 90 needs age 74.5 — and anyone that old already qualified at 67 under the age rule, which places them in the Incentive. Plan 2 stays legally eligible for the Bridge under G2; it should be scored at zero, or VRS reserves for a population that does not exist.

5. Ledger 1 — VRS pension. What the member pays, and what we cannot compute

The member payment is verified exactly: Plan 1 pays 12.50% of salary per year of service purchased, Plan 2 pays 10.74%. The actuarial-equivalent cost is **not** computable outside VRS, so it is written as k times the member payment. The trust-fund subsidy is therefore (k – 1) times the member payment.

Table 7 — trust-fund subsidy over the four-year pilot, at four values of k.

Case	Participants	Member payments	Subsidy at k = 1.5	Subsidy at k = 2.0	Subsidy at k = 2.5	Subsidy at k = 3.0
Low	9.1	$121k	$61k	$121k	$182k	$243k
Mid	42	$555k	$277k	$555k	$832k	$1.11M
High	141	$1.88M	$939k	$1.88M	$2.82M	$3.76M

Table 8 — the same thing per participant, and as a discount.

Case	k	Actuarial-equivalent value	Subsidy	Subsidy per participant	Share subsidised
Low	1.5	$182k	$61k	$6,682	33%
	2.0	$243k	$121k	$13,365	50%
	2.5	$303k	$182k	$20,047	60%
	3.0	$364k	$243k	$26,729	67%
Mid	1.5	$832k	$277k	$6,682	33%
	2.0	$1.11M	$555k	$13,365	50%
	2.5	$1.39M	$832k	$20,047	60%
	3.0	$1.66M	$1.11M	$26,729	67%
High	1.5	$2.82M	$939k	$6,682	33%
	2.0	$3.76M	$1.88M	$13,365	50%
	2.5	$4.70M	$2.82M	$20,047	60%
	3.0	$5.64M	$3.76M	$26,729	67%

k is the number that decides this bill and we do not have it. k will be high for this cohort. Actuarial cost rises sharply with age, and both pathways deliberately aim at the oldest members in the plan. This is the first item in the G24 data request.

Two further Ledger 1 lines are not computed here, deliberately. **Benefit acceleration** — the cost of a pension starting earlier than it otherwise would — cannot be estimated without VRS's own decrement assumptions, and it is the real cost driver for the Bridge. **True-up recoveries** under G5 reduce the subsidy by an unknown amount, because nobody knows how many participants will decline to retire. Both are listed as VRS calculations in the G24 request.

Table 9 — annual pension increase from the purchased service, for scale only. Not a liability figure.

Case	Participants	Added annual pension, all participants	Per participant	× 20 years, undiscounted
Low	9.1	$17k	$1,824	$331k
Mid	42	$76k	$1,824	$1.51M
High	141	$256k	$1,824	$5.13M

Table 9 is arithmetic on the benefit multiplier, not a present value. It is here so that nobody has to guess the order of magnitude, and it must not be quoted as a liability.

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6. Ledger 2 — employer and workforce. Agency appropriation, not trust-fund money

Two dials govern this ledger, and both are printed in full. **Replacement salary** is the share of the departing employee's salary that the replacement earns — 80%, 85% or 90%, that is 20% to 10% less. **Positions not refilled** is the share of vacated positions that are eliminated — 1%, 3% or 5%.

A position not refilled saves the whole package. A position refilled saves only the difference between the incumbent and the replacement. Both are scaled by 1.1223 for group life, administration and FICA.

Table 10 — annual recurring agency payroll saving, every combination. 27 cells.

Case	Not refilled	Replacement salary	Annual saving	— from unfilled posts	— from salary compression	Amortization timing cost
Low	1%	80%	$179k	$8,596	$170k	$10k
	1%	85%	$136k	$8,596	$128k	$7,879
	1%	90%	$94k	$8,596	$85k	$5,418
	3%	80%	$193k	$26k	$167k	$11k
	3%	85%	$151k	$26k	$125k	$8,724
	3%	90%	$109k	$26k	$83k	$6,313
	5%	80%	$206k	$43k	$163k	$12k
	5%	85%	$165k	$43k	$122k	$9,569
	5%	90%	$125k	$43k	$82k	$7,208
Mid	1%	80%	$817k	$39k	$778k	$47k
	1%	85%	$623k	$39k	$584k	$36k
	1%	90%	$428k	$39k	$389k	$25k
	3%	80%	$880k	$118k	$762k	$51k
	3%	85%	$690k	$118k	$572k	$40k
	3%	90%	$499k	$118k	$381k	$29k
	5%	80%	$943k	$196k	$747k	$55k
	5%	85%	$756k	$196k	$560k	$44k
	5%	90%	$570k	$196k	$373k	$33k
High	1%	80%	$2.77M	$133k	$2.63M	$160k
	1%	85%	$2.11M	$133k	$1.98M	$122k
	1%	90%	$1.45M	$133k	$1.32M	$84k
	3%	80%	$2.98M	$399k	$2.58M	$172k
	3%	85%	$2.33M	$399k	$1.94M	$135k
	3%	90%	$1.69M	$399k	$1.29M	$98k
	5%	80%	$3.19M	$665k	$2.53M	$185k
	5%	85%	$2.56M	$665k	$1.90M	$148k
	5%	90%	$1.93M	$665k	$1.26M	$112k

Read the two components separately. At 1% not refilled, almost the entire saving is salary compression; at 5%, unfilled posts carry about a third of it. Salary compression is the durable component, and it is the one an agency can actually plan around.

Do not let anyone count 11.07% as avoided employer cost. The State Employees employer contribution rate is 11.07%, but 6.49 of those points are amortization of a **fixed dollar** unfunded liability of $6.05 billion. Cutting payroll does not cut that liability — it cuts the dollars collected against it, and the rate rises later to make up the difference. Only 4.58 points plus FICA is a real saving. The amortization column above is therefore a **timing cost, not a saving**, and it is shown separately so that nobody can add it to the wrong side.

Table 11 — the mid-dial workforce case: 85% replacement salary, 3% of positions not refilled.

Case	Annual saving	Four-year total	Per participant per year	Amortization timing cost	Net of the timing cost
Low	$151k	$603k	$16,610	$8,724	$142k
Mid	$690k	$2.76M	$16,610	$40k	$650k
High	$2.33M	$9.34M	$16,610	$135k	$2.20M

7. The Bridge screen — why the pool is 2,396 and not 5,177

Purchased service adds **service**, not **age**. A member whose only route to unreduced retirement is Plan 1's age-65 rule or Plan 2's age-67 rule cannot be bridged at any price: four years of military credit moves their eligibility date by zero days. G9's screen — within four years of a threshold — counts those members. The purchasable screen does not.

Table 12 — the two screens compared.

Case	Screen	Bridge pool	Bridge participants	Total participants	Member payments
Low	Purchasable — **adopted**	2,396	2.5	9.1	$121k
	G9 as written — **retired**	5,177	5.4	12.0	$180k
Mid	Purchasable — **adopted**	2,396	11.5	42	$555k
	G9 as written — **retired**	5,177	25	55	$823k
High	Purchasable — **adopted**	2,396	39	141	$1.88M
	G9 as written — **retired**	5,177	84	186	$2.79M

The G9 screen overstates the Bridge pool by **2.2×** and total participants by about 24%. **Decision 3, taken 12 October 2026: the purchasable screen is adopted.** The G9 rows above are kept only so the difference can be audited; 5,177 is retired as a Bridge figure and may appear in a fiscal note only labelled as the superseded screen, with the reason. We chose the smaller defensible pool over the larger one, and the fiscal note says so rather than leaving it to be found.

8. Ledger 3 — net Commonwealth, and the one rule the model never breaks

The agency payroll saving is agency appropriation. It is not VRS trust-fund money, and nothing in the current design moves it into the trust fund. That is G11, and it holds in every table above. The two ledgers may be presented side by side as separate Commonwealth effects. They may never be netted in a headline, a fiscal note or an advocacy document.

Table 13 — the two ledgers side by side. Four-year pilot, 85% replacement salary, 3% not refilled, k = 2.0.

Case	Ledger 1 — member payments received	Ledger 1 — trust-fund subsidy at k = 2.0	Ledger 1 — acceleration cost (VRS)	Ledger 2 — agency saving per year	Ledger 2 — agency saving over four years
Low	$121k	$121k	Not computable here	$151k	$603k
Mid	$555k	$555k	Not computable here	$690k	$2.76M
High	$1.88M	$1.88M	Not computable here	$2.33M	$9.34M

Columns 2 and 3 are equal at k = 2.0 by definition: the subsidy is (k – 1) times the member payment. At k = 3.0 the subsidy doubles while the member payment does not move. See Table 7.

At the mid case the agency saving runs about $690,000 a year while the trust-fund subsidy over the whole four years is about $555,000 at k = 2.0. That comparison is tempting and it is exactly the comparison G11 forbids, because the money sits in different funds and no mechanism connects them.

Decision 6, taken 12 October 2026: the bill carries no transfer mechanism. The saving stays in agency appropriations permanently and is presented as a separate Commonwealth benefit. The question is closed. G22 still measures the saving every year, so reenactment in 2031 can revisit the point from Virginia data instead of from this model.

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9. Administrative cost, and what is still open

G14 requires administrative expense to be shown separately, and it is not estimated here. The V1 figure of $375,000 one-time was built for a different architecture — a certified-subsidy cap, a published queue and a central reservation system, none of which survive into V2. V2 adds work V1 did not have: a per-member calculation of the exact months needed to reach a threshold, a six-month retirement check, and a true-up assessment and collection process. A fresh estimate has to come from VRS, and it belongs in the G24 request rather than in this model.

Table 14 — what is still open, in the order each one changes the numbers. Four of the seven questions in the first version of this table were board decisions. All four were taken on 12 October 2026 and are shown as closed.

#	Question	Owner	What it moves
1	k — the ratio of actuarial-equivalent cost to approximate normal cost, by age band	VRS	Everything in Ledger 1. Currently printed as a range, not an answer.
2	Is the G5 true-up actually collectible?	VRS counsel	Whether the Incentive pathway has any hard tie to a retirement outcome at all. The Incentive cohort is already retirement-eligible, so the purchase raises the benefit and accelerates nothing. The likely answer is to hold the credit off the member's record until either retirement or full actuarial payment.
3	Actual count of Plan 1 / Plan 2 state employees, veterans among them, and those with qualifying purchasable service	VRS and DHRM	Replaces the 28,994 derivation and closes two of the three dials in Table 2.
—	✅ **Closed** — which screen defines the Bridge population	Board	**Decided: the purchasable screen, 2,396.** Worth about 24% of total participants. Section 7
—	✅ **Closed** — do the hazardous-duty and judicial state plans take part	Board	**Decided: excluded, expressly in the bill.** SPORS 1,902, VaLORS 7,459, JRS 473. Their approximate-normal-cost price rates run roughly 1.9× and 2.6× the non-hazardous rate, their eligibility rules differ, and we hold no veteran count for any of them. Expect a floor amendment to add VaLORS; pre-score it
—	✅ **Closed** — does the bill carry a mechanism to move the agency saving into the trust fund	Board	**Decided: no mechanism.** Nothing in Ledger 2 ever reaches Ledger 1. Section 8
—	✅ **Closed** — the Plan 2 findings	Board	**Decided: both confirmed.** Plan 2 Bridge scored at zero for the whole pilot; the 12% and 13% proximity assumptions withdrawn in favour of the derived 4.0% and 6.7%
4	A V2 administrative-cost estimate	VRS	G14. Shown separately, never inside the pension estimate.

Sources

VRS, *Annual Actuarial Valuation of the Virginia Retirement System, June 30, 2025* — active participant detail, State Employees, p. 50. · VRS, approximate-normal-cost rate chart effective 1 July 2025. · VRS, *Purchase of Prior Service* — eligibility, types, payment methods. · VRS, *Introduction to VRS* brochure, p. 4 — Plan 1 110,474, Plan 2 67,333, combined 177,807 at 30 June 2025, all 864 participating employers. · Virginia Code § 51.1-142.2 and § 51.1-155. All retrieved 12 October 2026.

Figures in this document were computed by a port of the interactive model's own code. Every headline figure the model reports — 17 / 42 / 86 on the board's take-up bounds, the 9-to-141 full range, member payments of $231k / $555k / $1.16M — reproduces exactly.